Alif Tasbir

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PROFILE

MSc Artificial Intelligence candidate at De Montfort University who builds machine learning systems end to end, from deep learning research to deployed applications. Experienced in deep learning for EEG and ECG signals, LLM fine-tuning, retrieval-augmented generation (RAG), multi-agent systems and efficient edge inference, with three peer-reviewed papers in clinical AI. Seeking an AI/ML Engineer role building machine learning solutions for real-world problems.

EDUCATION

MSc Artificial Intelligence, De Montfort University, Leicester, UK | Sep 2025 to Sep 2026 (expected)
Modules: Neural Systems and Natural Language Processing; Fuzzy Logic and Evolutionary Computing.

BSc Computer Science and Engineering, University of Information Technology and Sciences (UITS), Dhaka, Bangladesh | Aug 2018 to Aug 2022
CGPA 3.74/4.00. Modules: Machine Learning, Data Structures and Algorithms, Software Engineering, Object Oriented Programming.

TECHNICAL SKILLS

- **Languages:** Python, C, C++, Java, JavaScript, HTML, CSS
- **Machine learning and deep learning:** PyTorch, TensorFlow, Keras, scikit-learn, Pandas, NumPy, CNNs, LSTMs
- **LLMs and RAG:** LLM fine-tuning (LoRA, QLoRA, PEFT, TRL), Hugging Face Transformers, retrieval-augmented generation (RAG), hybrid search and reranking, LangChain, LangGraph agents, Model Context Protocol (MCP), Ollama, Sentence Transformers, vector databases (FAISS, ChromaDB), LLM evaluation (EM/F1, faithfulness, significance testing, BLEU, ROUGE, BERTScore), Gemini API, natural language processing (NLP)
- **APIs and apps:** FastAPI, Flask, Django, Streamlit, Gradio
- **MLOps and tools:** Git, GitHub Actions (CI), MLflow, DVC, Docker, OpenTelemetry
- **Data:** SQL (MySQL), NoSQL, Excel

SELECTED PROJECTS

Fine-Tuned Small LLM for Multi-Hop RAG | Python, PyTorch, PEFT, FAISS, LangGraph, MCP, FastAPI, OpenTelemetry
- Fine-tuned Qwen3-1.7B with LoRA for multi-hop question answering on HotpotQA, raising exact match from 10.4 to 58.4 and matching a 4.7× larger Qwen3-8B (71.2 vs 69.3 F1) with ~5× smaller weights at ~8× the throughput.
- Built hybrid retrieval (BM25 + FAISS dense search, rank fusion, cross-encoder reranking) with iterative multi-hop search, raising evidence recall from 54% to 89%; trained the model to cite sources and abstain without evidence.
- Exposed it as an MCP server with typed tools and OpenTelemetry tracing, fine-tuned a LangGraph agent for tool use (+19.7 F1), and made training 2.8× faster with 55% less GPU memory.

Multimodal RAG Application | Python, FastAPI, Streamlit, FAISS, Gemini 2.5 Flash, gpt-oss-20b
- Built a bilingual (Bangla and English) RAG system on a custom FAISS vector store for semantic retrieval.
- Designed a FastAPI backend supporting real-time web search and an adaptive deep-reasoning mode.
- Added dynamic routing between Gemini 2.5 Flash and gpt-oss-20b to balance answer quality against API rate limits.

Heterogeneous Multi-Agent Edge AI for Clinical Decision Support (MSc dissertation)
- Built a two-agent ECG pipeline: a CNN-LSTM heartbeat classifier sends events to a frozen Gemma 4 E4B model, either as text or through a trained linear adapter that emits 4 virtual tokens per event.
- Cut the language model's prompt-processing time by 24–78% against compact text, and improved balanced accuracy at 10 and 20 events (0.83 vs 0.73), with held-out patients, three training seeds and patient-level bootstrap statistics.
- Ran batched serving on a 12 GB GPU: 10–30% higher median throughput and 12–28% lower median energy per decision than filtered text. Measured energy with NVML power sampling at 100 Hz.

WORK EXPERIENCE

Junior Filing Specialist, Compliance Services Limited (part of WSD Group), Dhaka, Bangladesh | Apr 2024 to Sep 2025
- Handled live filings and manually proofread documents to verify accuracy.
- Converted Word documents to HTML and built webpages using HTML and CSS.
- Maintained filing records in Excel, kept documentation up to date and analysed data.

RESEARCH EXPERIENCE

Research Assistant, University of Information Technology and Sciences (UITS), Dhaka, Bangladesh | Jul 2022 to Mar 2023
- Developed graph-based deep learning models for epileptic seizure detection from EEG signals, improving detection accuracy.
- Applied signal processing, CNNs and statistical inference to optimise model results.
- Contributed to grant proposals and experiment design.

PUBLICATIONS

1. Parkinson's disease detection from EEG signal employing autoencoder and RBFNN-based hybrid deep learning framework utilizing power spectral density. *Digital Health*, SAGE, 2024.
2. Graph attention network and radial basis function neural network-based hybrid framework for epileptic seizure detection from EEG signal. *Journal of Autonomous Intelligence*, Vol. 7, Issue 3, 2023.
3. Epileptic seizure detection from EEG signals using linear graph convolutional network and DenseNet based hybrid framework. *Journal of Radiation Research and Applied Sciences*, Elsevier, Vol. 16, 2023.